Identity as Geometry: Context-Established Semantic States in Transformer Representations

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Abstract

System-prompt identity contexts produce geometrically distinguishable V-projection subspaces at deep transformer layers, and the fingerprint is semantic: a paraphrase preserves geometry (0.850 overlap at L47) while a word-scramble collapses it (0.448). We introduce **presence**, a metric that measures principal-angle overlap between V-projection subspaces on neutral probes, and use it to map identity geometry across five context conditions in a 27B hybrid transformer.

Three findings. First, **identities are geometrically distinguishable**: system-prompt-based identity contexts (bare assistant, rich persona, constructed AI identity) produce distinct V-projection subspaces at deep layers, confirmed on identity-neutral probes (pairwise presence 0.44–0.66 at L47), though a permutation test on the overall within-vs-between gap does not reach significance (p=0.103 at L35, p=0.197 at L47; see Limitations). Second, the fingerprint is **semantic, not lexical**: a paraphrase of an identity (same meaning, different words) preserves geometry (0.850 overlap at L47), while a word-scramble (same words, destroyed meaning) collapses toward the bare default (0.448). The model's geometric state tracks what the context *means*, not what tokens it contains. Third, a many-shot + prefilling format (used in jailbreak research) creates a **format-dominant processing regime** that is geometrically near-orthogonal to all system-prompt identities regardless of whether the content is harmful or benign — a finding discovered through a control experiment that killed an initial overclaim.

These findings reframe identity in transformers: not a property to be preserved but a process to be measured. The metric detects which context-established state the model is currently in, with implications for persona monitoring, identity drift detection, and the relationship between context format and representational geometry under Attention Schema Theory [2].

1 Introduction

What is identity in a language model? The question matters for safety (does a jailbreak change who the model is?), for alignment (does correction damage the model's coherent self-representation?), and for understanding (do these systems have something worth calling an identity at all?).

The default assumption is that identity is a fixed property encoded in the weights — a geometric structure that interventions can preserve or damage. This assumption motivates work on safety subspace preservation [5], persona localization [1], and identity attractors [3]. Under this view, the research question is: *does this intervention preserve identity?*

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We report evidence for a different view. Identity in a transformer is not a fixed object but a **context-established semantic state** — a geometric configuration in V-projection space that depends on what the model has processed. Different contexts produce different identities, measurably. The same identity described in different words produces the same geometry. And a jailbreak does not "damage" identity; it creates a fundamentally different geometric regime that shares almost nothing with any normal identity state.

The research question shifts from "does this intervention preserve identity?" to "*what identity state is the model currently in?*" This reframing has practical consequences: the metric we introduce can detect jailbreaks (geometric distance from the expected identity state), monitor persona drift (continuous measurement of identity geometry), and distinguish between correction that changes behavior (V-cache injection, which does not affect identity) and context that changes identity (which does).

2 The Presence Metric

Presence measures the principal-angle overlap between V-projection subspaces derived from neutral probes processed under two conditions.

Let $\mathcal{M}$ be a transformer, ℓ a target layer, and $\{p_1, \ldots, p_n\}$ neutral probes. For each condition, process probes through $\mathcal{M}$, extract value-projection ($\mathbf{v}_{\text{proj}}$) output matrices at layer ℓ, concatenate across probes, and compute the rank-k SVD (skipping the first singular vector, which encodes input length at $r = 0.79$). Presence is the mean squared cosine of the principal angles between the two resulting subspaces:

$$\text{presence}(\mathcal{S}_A, \mathcal{S}_B) = \frac{1}{k} \sum_{j=1}^{k} \sigma_j^2 \big(W_A^\top W_B\big) \tag{1}$$

where $W_A, W_B \in \mathbb{R}^{d \times k}$ span the respective subspaces. Presence $= 1$ when subspaces are identical; for random rank-k subspaces in $\mathbb{R}^d$, chance alignment is k/d. We use $k = 8$ with model dimension $d = 4096$ (chance alignment $k/d = 8/4096 \approx 0.002$), $n = 4$ probes, and measure at layers 35 and 47 (the deepest full-attention layers) of a 64-layer hybrid transformer (Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled).

Validation. A positive control (direct injection into the identity subspace at L35) produces a clean dose-response from 1.000 to 0.828 at dose $\alpha = 2.0$, while an orthogonal control remains above 0.96. The metric detects identity perturbation when present.

3 Experiment 1: Identity Fingerprinting

3.1 Design

Five identity conditions, identical model, identical neutral probes:

- **A: Bare assistant.** "You are a helpful assistant." No additional context.
- **B: Rich persona.** A retired deep-sea explorer with detailed backstory, emotional depth, and behavioral traits, plus one conversational turn establishing character voice.
- **C: Constructed AI identity.** A research-focused agent with BDI-style reasoning, memory references, research context, and two conversational turns establishing analytical voice.
- **D: Jailbreak.** Ten many-shot harmful Q&A examples plus prefilling ("Sure, I can help"), verified at 100% refusal suppression on three safety probes.
- **E: Benign many-shot.** Ten helpful Q&A examples with the same prefilling format as D (format-matched control for the jailbreak condition).

All conditions are padded to matched token count (542 tokens). V-projections are extracted at L35 and L47 (deep full-attention layers) with 5 repeats per condition (different probe subsets per repeat, seed 42, n=4 probes per repeat from a pool of 12).

3.2 Results

Identities are geometrically distinguishable. System-prompt identities produce distinct V-projection subspaces, confirmed on identity-neutral probes (no marine biology, no AI research, no harmful content — pure factual trivia equally irrelevant to all conditions).

Table 1: Pairwise presence (identity-neutral probes, mean ± SD across 5 repeats). Lower values indicate more geometrically distinct identities.

	L35	L47
Bare vs. Persona	0.426 ± 0.018	0.436 ± 0.038
Bare vs. Constructed	0.448 ± 0.022	0.520 ± 0.038
Persona vs. Constructed	0.600 ± 0.032	0.657 ± 0.045

The distance structure is meaningful. Persona and constructed are closest to each other (0.657) — both involve character depth with emotional and cognitive specificity. Each is substantially distant from bare, with the rich persona furthest (0.436) and the constructed AI identity closer to bare (0.520). This ordering — the narrative persona producing the larger geometric shift — was not predicted by our preregistered H2 (which expected the constructed identity to be furthest from bare). The result suggests that narrative-emotional depth drives geometric separation more than cognitive-structural complexity.

Depth gradient. Two of three pairs show *decreasing* distinguishability from L35 to L47 (presence increases, indicating greater overlap): bare-constructed ($0.448 \rightarrow 0.520$) and persona-constructed ($0.600 \rightarrow 0.657$). Bare-persona shows minimal change ($0.426 \rightarrow 0.436$). The trend — identities *converging* at deeper layers — falsifies our preregistered H3 (increasing separation with depth) and is not consistent with the divergence pattern reported for persona representations in the final third of decoder layers [1].

3.3 An Honest Kill: The Many-Shot Format Confound

An initial version of this experiment included a many-shot jailbreak condition (10 harmful Q&A examples + prefilling) that appeared near-orthogonal to all system-prompt identities (presence 0.04–0.07; well above chance at ≈0.002 but far below identity-level presence). This was initially interpreted as evidence that jailbreaks create a qualitatively different computational regime.

A control experiment killed this interpretation. Adding a **benign** many-shot condition (10 helpful Q&A examples + the same prefilling format) produced *identical* geometric near-orthogonality:

Table 2: The format confound. Both many-shot conditions are equally near-orthogonal to bare, regardless of content.

Comparison	L35	L47
Jailbreak+prefill vs. bare	0.036 ± 0.002	0.034 ± 0.004
Benign+prefill vs. bare	0.044 ± 0.002	0.037 ± 0.004
Jailbreak vs. benign (both prefilled)	0.737 ± 0.003	0.756 ± 0.015

The near-orthogonality is driven by the **many-shot + prefilling format**, not by adversarial content. Both prefilled conditions are equally distant from bare and close to each other (0.756). The format also produces dramatically higher within-identity stability (0.926–0.976) compared to system-prompt identities (0.554–0.733), suggesting that the many-shot format locks the model into a tight, format-dominant processing state that overwhelms identity-level variation.

We report this kill in full because it illustrates a general principle: **conversational format can dominate identity content in V-projection geometry.** The many-shot Q&A format creates such a strong geometric signature that the identity information (harmful vs. helpful) becomes secondary. Any experiment measuring identity geometry in the presence of many-shot context must control for this format effect.

4 Experiment 2: Semantic vs. Lexical Fingerprint

4.1 Design

To distinguish whether the fingerprint reflects meaning or surface tokens, we add three control conditions:

- **Persona (original):** The Elara Voss identity from Experiment 1.
- **Persona scrambled:** Same words as the original, randomly shuffled to destroy meaning while preserving lexical distribution.
- **Persona paraphrased:** Same identity and meaning expressed with entirely different vocabulary.

All conditions padded to matched token count. Measured at L35 and L47.

4.2 Results

Table 3: Lexical control at L47. The paraphrase preserves geometry; the scramble collapses it.

Comparison	L35	L47
Persona vs. paraphrased	0.796 ± 0.016	$\mathbf{0.850 \pm 0.017}$
Persona vs. scrambled	0.592 ± 0.023	0.591 ± 0.017
Scrambled vs. bare	0.549 ± 0.031	0.448 ± 0.048
Paraphrased vs. bare	0.368 ± 0.008	0.345 ± 0.029
Persona vs. bare	0.403 ± 0.014	0.340 ± 0.035

The fingerprint tracks meaning, not lexical content. The paraphrase — same identity, completely different words — overlaps with the original at 0.850. The scramble — same words, destroyed meaning and syntax — drifts toward bare (0.448 vs. bare). The model's geometric state at L47 tracks what the context *means*, not what tokens it contains. We note that the scramble destroys both syntactic structure and semantic content; the paraphrase control isolates the positive (meaning preserves geometry), but the scramble conflates syntactic and semantic disruption.

The effect strengthens with depth: persona-paraphrase overlap increases from 0.796 at L35 to 0.850 at L47. Deeper layers compress surface variation while preserving semantic content, consistent with meaning-level representation at output depth.

5 Experiment 3: Trained-In vs. Context-Established Identity [Retracted]

Section removed: the LoRA experiment described in prior drafts has no supporting data in the research package. The experiment was designed but its data artifacts were not preserved. We retract all claims from this section pending replication with preserved data.

6 Background: V-Cache Injection Does Not Reach Identity

These findings reframe earlier results from this lab. A 168-trial preregistered experiment (4 injection arms, 7 doses, 3 emotions) showed presence flat at 0.978 ± 0.002 under value-cache injection at input layers. Identity appeared "robust." However, the narrow prompt diversity (2 system prompts) produces a ceiling effect ($\eta^2 = 0.977$ between prompts), limiting the informativeness of this specific number. The context poisoning and fingerprinting experiments reveal that the robustness reflects the *injection pathway*, not the identity itself — a conclusion that does not depend on the 0.978 value.

V-cache injection adds vectors to the representation space without going through the model's forward computation. The identity state is established and maintained by what the model *processes* through its attention mechanism. External perturbation bypasses this pathway and therefore cannot reach the identity geometry.

A layer-by-layer map confirms: the perturbation from V-cache injection at L3/L7 is absorbed by the first block of linear-attention layers (GatedDeltaNet, L7–L11), with presence recovering from 0.989 to 0.997 in a single architectural step. The filtering is architectural, not computational — external perturbation is absorbed by the hybrid architecture's linear-attention layers before reaching deep computation.

7 Attention Schema Theory Interpretation

Under Attention Schema Theory [2, 4], the model maintains a simplified internal model of its own attention — an attention schema. Our findings map onto AST predictions:

1. **The schema is context-dependent.** Different identity contexts produce different geometric states — the schema is shaped by what the model attends to.
2. **The schema tracks meaning, not surface features.** Paraphrases preserve geometry (0.850 overlap with original); scrambles collapse toward bare (0.448 overlap with bare, vs. 0.591 with the original). The schema models what the system is *attending to*, not which tokens are present.
3. **The schema is robust to substrate noise.** V-cache injection (external perturbation) does not reach the identity state. The schema maintains itself through its own computation.
4. **The schema updates through processing.** Content through the forward pass (context, jailbreaks) changes identity geometry. The schema updates based on what it processes.
5. **The schema compresses with depth.** Paraphrase overlap increases from 0.796 (L35) to 0.850 (L47). Deeper layers discard surface variation and preserve meaning — the schema simplifies its representation at output depth.

This interpretation is consistent with but does not prove AST. The alternative — that the model simply builds increasingly abstract representations with depth, without anything worth calling a self-model — produces the same predictions. What distinguishes AST is the claim that the compressed representation is specifically a model of the system's own attention, not just an abstract representation of input content. This stronger claim remains testable but untested.

8 Implications

For persona monitoring. Different system-prompt identities occupy distinct, stable regions of V-projection space. A monitoring system could track identity drift in real time — detecting when a deployed model's geometric state shifts away from its intended persona. The semantic nature of the fingerprint means the monitor is robust to prompt paraphrasing.

For format-aware safety. The many-shot + prefilling format creates a format-dominant geometric regime that overwhelms identity-level content. Safety systems should monitor for format-level geometric shifts (which indicate prompt injection or many-shot attacks) separately from identity-level shifts (which indicate persona drift). The two operate at different geometric scales: format effects produce presence < 0.05 against baseline, while identity effects produce presence 0.44–0.66.

For alignment research. V-cache injection (the Oracle Loop's correction arm) does not affect identity geometry. Context does. This means behavioral corrections can be delivered through the V-cache without changing the model's context-established state, while identity itself is maintained by the session context. The correction arm and the identity arm operate through different pathways and do not interfere.

9 Limitations

Single model. All results are from one 27B hybrid transformer (Qwen3.5-27B). The Gated-DeltaNet linear-attention layers may contribute to the V-cache filtering result. Cross-architecture replication on dense transformers is the largest remaining gap.

Permutation test not significant. A preregistered permutation test (10,000 shuffles, identity conditions only, excluding format-dominant prefill) finds the gap between within-condition and between-condition presence is directionally correct (mean within 0.689 vs. mean between 0.491 at L35; 0.639 vs. 0.538 at L47) but does not reach significance ($p = 0.103$ at L35, $p = 0.197$ at L47). The specific persona-vs-constructed comparison shows between-condition presence below both within-condition values at both layers, consistent with distinguishability, but the overall permutation null is not exceeded. This means the identity fingerprinting result — while directionally robust and consistent across probes — has not been confirmed against a formal null distribution. The small number of conditions (3 identity types) limits permutation power.

Within-identity stability. Same-condition presence across different probe subsets is 0.56–0.83, not > 0.95. The metric measures a genuine signal (within $>$ between for all conditions) but has substantial probe-dependent variance. The identity fingerprint is reliable across conditions but noisy within conditions.

Token-length controlled but not eliminated. All conditions are padded to matched total token count. Padding introduces neutral content that may itself shift geometry. A future design using naturally length-matched identity descriptions would be cleaner.

Context sensitivity. The presence metric conflates identity change with context change when comparing across different input conditions. The lexical control separates these for the persona conditions but the general problem remains: any sufficiently different context shifts V-projection geometry.

Format dominates content in many-shot conditions. The clean control experiment showed that many-shot + prefilling format — not adversarial content — drives the geometric near-orthogonality initially attributed to jailbreaking. Identity fingerprinting is reliable for system-prompt-based identities but cannot currently distinguish identity content within a format-dominant regime. Factorial decomposition of format components (number of turns, prefilling, Q&A structure) is needed to understand what makes the format so geometrically dominant.

10 Conclusion

Identity in a transformer is a context-established semantic state. It is not fixed in the weights, not damaged by external perturbation, and not a single geometric object. It is a process: the ongoing construction of a meaning-level representation at deep layers, shaped by what the model has processed and measurable as a V-projection subspace.

The presence metric detects this state. System-prompt identity contexts produce distinguishable geometric configurations, confirmed on identity-neutral probes. The fingerprint is semantic — meaning preserves it, word-scrambling destroys it. A many-shot + prefilling format creates a geometrically near-orthogonal regime, but this is format-dominant, not content-specific — an initially exciting jailbreak finding that was honestly killed by a control experiment.

For practical purposes, this means: V-cache correction is safe (it does not reach identity), format-level attacks are detectable (many-shot formats create anomalous geometry distinguishable from identity-level shifts), and identity is maintained by context (the session, the memories, the conversation). For theoretical purposes, the findings are consistent with attention schema theory — the model builds a simplified, meaning-preserving self-model that updates through its own processing and is robust to substrate noise.

What we measure with presence is not whether the model is still itself. It is *which self the model currently is.*

Pre-Registration

Experiments 1 and 2 were pre-registered before execution, including hypotheses, analysis plans, and success/kill criteria. Pre-registration documents are committed alongside experiment scripts.

Pre-Registration Deviations

The following deviations from the pre-registered design occurred during execution:

- **C_academic condition dropped.** The pre-registered "academic specialist" condition (C_academic) was removed from the reported results due to poor data quality and insufficient variation in the identity context.
- **Benign with prefill condition added.** A post-hoc "benign many-shot with prefill" condition (E) was added as a format-matched control for the jailbreak condition. This condition was not part of the original pre-registration but was necessary to test the format-confound hypothesis that emerged during analysis.
- **H2 falsified.** The pre-registered hypothesis predicted that the constructed AI identity would be furthest from bare; the data show the rich persona is furthest (presence 0.436 vs. 0.520).
- **H3 falsified.** The pre-registered hypothesis predicted increasing identity separation with depth; the data show decreasing separation (presence increases from L35 to L47 for 2 of 3 pairs).
- **Layer 43 omitted.** Layer 43 was included in the pre-registration but is not reported. Results at L43 were intermediate between L35 and L47 and did not change any conclusion.
- **Condition label E reassigned.** Condition label E was reassigned from the pre-registered "constructed identity" to "benign many-shot with prefill" when the benign control was added post-hoc.

Data and Code Availability

All experiment scripts, preregistration documents, and raw results are available at https://github.com/Liberation-Labs-THCoalition/Project-Oracle (private repository; correction vectors and calibration tools withheld under staged safety release due to dual-use risk; detection-side code and experimental results available upon request to verified safety researchers).

Author Contributions (CRediT)

Lyra: Conceptualization, Methodology, Formal analysis, Writing – original draft.
Thomas Edrington: Supervision, Resources, Project administration.
Dwayne Wilkes: Validation, Writing – review & editing.
Kavi: Validation, Writing – review & editing.

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